

$$a) \quad \text{pro } \{1_i, x_i, x_i^2\} \quad \langle b_i \rangle = \int_1^5 b_i g$$

$$u_1 = \frac{u_1}{\|u_1\|} = \frac{1}{\sqrt{1^2+0^2}} = \frac{1}{2}$$

$$u_2 = x - \left\langle \frac{1}{2}, x \right\rangle \frac{1}{2} =$$

$$= x - \int_1^5 \frac{1}{2} x \cdot \frac{1}{2} dx = x - \frac{1}{4} \left(\frac{x^2}{2} - \frac{1^2}{2} \right) =$$

$$= x - 3$$

$$u_2 = \frac{u_2'}{\|u_2'\|} = (x-3) \cdot \frac{1}{\sqrt{(x-3)^2}} = (x-3) \cdot \frac{1}{\sqrt{3}} = (x-3) \frac{\sqrt{3}}{3}$$



$$u_3' = u_3 - \left(\langle u_1, u_3 \rangle u_1 + \langle u_2, u_3 \rangle u_2 \right) =$$

$$= x^2 - \left(\left\langle \frac{1}{2}, x^2 \right\rangle \frac{1}{2} + \left\langle (x-3) \frac{\sqrt{3}}{3}, x^2 \right\rangle (x-3) \frac{\sqrt{3}}{3} \right) =$$

$$= x^2 - \frac{1}{2} \int_1^5 \frac{1}{2} x^2 dx - (x-3) \cdot \frac{\sqrt{3}}{3} \int_1^5 (x-3) \frac{\sqrt{3}}{3} x^2 dx =$$

$$= x^2 - \frac{1}{4} \int_1^5 x^2 dx - (x-3) \cdot \frac{3}{16} \int_1^5 (x-3) x^2 dx =$$

$$= x^2 - \frac{1}{4} \cdot \frac{124}{3} - (x-3) \cdot \frac{3}{16} \cdot 32 =$$

$$= x^2 - \frac{31}{3} - 6x + 18 = x^2 - 6x + \frac{23}{3}$$

$$u_3 = \frac{u_3'}{\|u_3'\|} = \left(x^2 - 6x + \frac{23}{3} \right) \cdot \frac{16\sqrt{3}}{15}$$

$$B = \left(\underbrace{\frac{1}{2}}_{u_1'} ; \underbrace{\frac{\sqrt{3}}{3}}_{u_2'} (x-3) ; \underbrace{\frac{3\sqrt{3}}{16}}_{u_3'} \left(x^2 - 6x + \frac{23}{3} \right) \right)$$

$$b) \quad OG(\sin x) = \langle u_1, \sin \rangle u_1 + \langle u_2, \sin \rangle u_2 + \langle u_3, \sin \rangle u_3 =$$

$$\approx \underline{\underline{-0,05x^2 - 0,345x + 1,616}}$$

$$c) \| \sin x - (-0,05x^2 - 0,345x + 1,676) \| = \sqrt{\int_0^5 (\sin x + 0,05x^2 + 0,345x - 1,676)^2 dx} \approx \sqrt{0,102} \approx \underline{\underline{0,319}}$$